Models with Non-Independent Errors

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Assumptions so far…

$$Data = Model + Error$$

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} \ldots + \beta_n x_{ni} + \varepsilon_i$$

- In these models, we have made important assumptions about the residuals (ε's).
  1. They are normally distributed. ***(normality assumption)***
  2. They have a constant variance. ***(homoscedasticity/homogeneity of variance assumption)***
  3. They are independent of each other. ***(independence assumption)***

Types of non-independence.

- ***Positive non-independence***: linked data are more alike than non-linked data.

Proficiency on a mathematical aptitude test:

School	Calculus I	College Algebra
1	660	510
2	740	560
3	780	650
4	650	500

- ***Negative non-independence***: linked data are more different than non-linked data.

Proficiency versus Growth on a mathematical aptitude test:

School	End Term Performance	Change over the Term
1	585	100
2	650	50
3	715	30
4	575	110

Sources of non-independence.

- In practice, non-independence is likely to arise to a surfeit of factors:
  - E.g., time within a person, kids within schools, patients with therapists, hospitals within counties, etc.

- Broadly speaking, we can think about these different sources of dependence as either ***grouped*** data or ***sequential*** data.

Team A

	Year
Player A	1, 2, 3, 4
Player B	1, 2, 3, 4
Player C	1, 2, 3, 4
…	

Team B

	Year
Player D	1, 2, 3, 4
Player E	1, 2, 3, 4
Player F	1, 2, 3, 4
…	

Team C

	Year
Player G	1, 2, 3, 4
Player H	1, 2, 3, 4
Player I	1, 2, 3, 4
…	

Returning to our examples…

- ***Positive non-independence***: linked data are more alike than non-linked data.

Proficiency on a mathematical aptitude test as a function of Class:

School	Calculus I	College Algebra	Resid_calc	Resid_alg
1	660	510	-47.5	-45
2	740	560	32.5	5
3	780	650	72.5	95
4	650	500	-57.5	-55
	$\bar{x}_{calc} = 707.5$	$\bar{x}_{alg} = 555.0$		

$$\hat{y}_i = \beta_0 + \beta_1 class_i$$

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If we ignore the fact that these classes are nested within schools, we violate a major assumption!

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- ***Negative non-independence***: linked data are more different than non-linked data.

Proficiency versus Growth on a mathematical aptitude test:

School	End Term Proficiency	Change over the Term	resid_prof	resid_growth
1	585	100	-46.25	27.5
2	650	50	18.75	-22.5
3	715	30	83.75	-42.5
4	575	110	-56.25	37.5
	$\bar{x}_{prof} = 631.25$	$\bar{x}_{grw} = 72.50$		

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If we ignore the fact that these classes are nested within schools, we violate a major assumption!

How do we handle the violation of this assumption?

- Control the data so that we no longer dealing with non-independent observation.
  - I.e., if scores from the same school are more likely to be like each other, we need to remove the ***variation due to school***, then we can get a clean look at variation due to ***class*** or ***proficiency versus growth***.

- The exact way that we do this will depend on whether our data are crossed or nested.

Definitions

- Two factors are ***crossed*** when every level of one factor co-occurs in the design with every level of the other factor.
  - I.e., there is at least one observation in every combination of levels for the two factors.

- A factor is ***nested*** within another factor when each category of the first factor co-occurs with only one category of the other.
  - I.e., an observation has to be within one level of Factor 2 in order to have a specific level of Factor 1. Not all combinations of levels are represented.

Definitions

- In many experimental paradigms, we refer to ***crossed*** factors as "within-subject" and ***nested*** factors as "between-subjects".
  - This nomenclature is valid, but understanding crossed versus nested as terms will help us understand more complicated statistical models later.

- To illustrate, consider two different training paradigms for changing body composition (%BF).

Definitions

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Participants at different levels:

	Time Point	
Group	Pre	Post
HIIT	*10*	*10*
Aerobic	*10*	*10*

Time and Group are crossed factors (every level of group is represented at every level of time).

Participants at different levels:

	Time Point	
Subject	Pre	Post
1	1	1
2	1	1
3	1	1
4	1	1
5	1	1
…		
17	1	1
18	1	1
19	1	1
20	1	1

Time and Subject are also crossed factors (every level of Subject is represented at every level of Time).

Definitions

- To illustrate, consider two different training paradigms for changing body composition (%BF)

Participants at different levels:

	Group	
Subject	HIIT	Aerobic
1	2	0
2	2	0
…		
9	2	0
10	2	0
11	0	2
12	0	2
…		
19	0	2
20	0	2

Subject, however, is nested within group, because only some subjects are represented in the HIIT group (1-10) and some are in the Aerobic group (11-20).

(I.e., Only some levels of Subject are represented at every level of Time).

Moving ahead:

- In the next two modules, we will demonstrate how to deal with non-independence
  - in ***nested*** designs (e.g., students within classrooms),
  - in ***crossed*** designs (e.g., participants exposed to different experimental conditions),
  - and then in situations where we are ***mixing*** crossed and nested variables.